

Vendor Sales & Profitability Analysis

Chandru · Python · SQLite · MySQL · Power BI · 15M+ Records





Business Problem

Fragmented Data

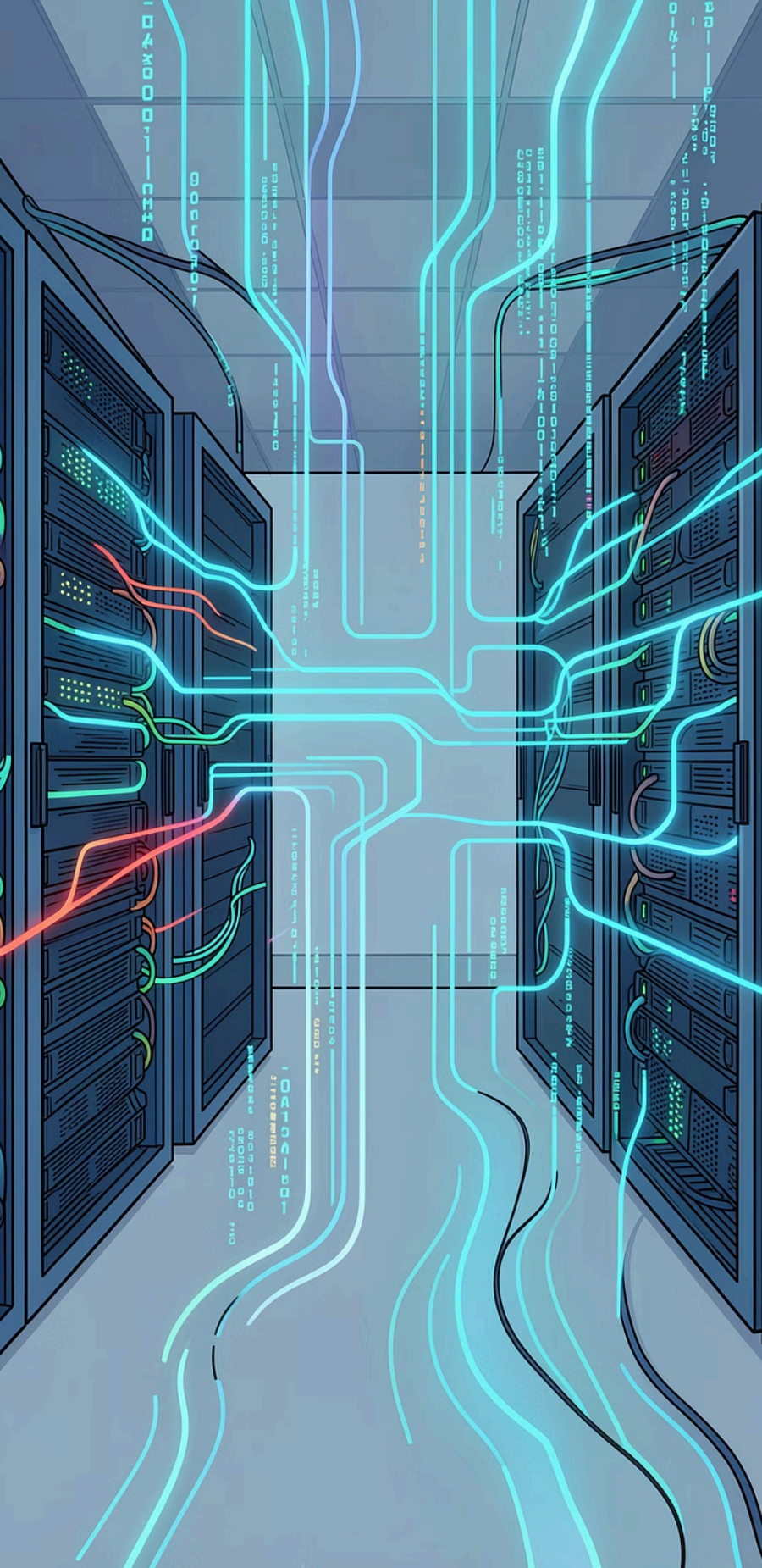
Purchases, sales, inventory & pricing across multiple tables

Key Questions

Top vendors? Profitability?
Revenue concentration? Freight impact?

Goal

Support vendor selection, pricing strategy & inventory decisions



Dataset Overview

15M+

Total Records

Across all operational tables

12.8M

Sales Records

Largest transaction table

2.37M

Purchase Records

Vendor purchase transactions

10.7K

Summary Records

Aggregated vendor_sales_summary

Data Engineering Pipeline



Ingest CSVs

Transform

Store

Analyze

Python pipeline automates ingestion, transformation, and storage — **vendor_sales_summary** becomes the single source of truth for dashboarding.

Data Cleaning & Feature Engineering

Data Cleaning

- Volume → numeric format
- Nulls → defaults
- Trimmed vendor & product names

Derived Metrics

Gross Profit

Sales Revenue – Purchase Cost

Profit Margin %

Gross Profit / Revenue × 100

Stock Turnover

Sales Qty / Purchase Qty

Revenue, gross profit & profitability ranking



80/20 rule — revenue concentration across vendors



Freight cost analysis on vendor profitability



Top products by revenue & gross profit

Sqibb

C

Made with **GAMMA**

Power BI Dashboard



KPI Section

Revenue · Gross
Profit · Margin · Sales
Qty · Active Vendors

Vendor Analysis

Top 10 by revenue &
profit · Scatter:
revenue vs
profitability

Product Analysis

Top 10 products by
revenue & gross
profit

Filters

Vendor · Product · Brand — dynamic exploration



Key Findings



Top Vendors

DIAGEO, Martignetti, Pernod Ricard, Bacardi, Brown-Forman lead revenue & profit



Top Products

Jack Daniels No. 7, Tito's Vodka, Absolut, Captain Morgan, Ketel One



Revenue Concentration

Pareto confirms small vendor group drives majority of revenue



Margin Gaps

High-revenue products show lower margins — pricing opportunity

Business Recommendations



Vendor Management

Strengthen top vendors; review profitability, not just revenue



Pricing Optimization

Reassess markup on high-sales, low-margin products



Freight Cost Control

Negotiate logistics; add freight-adjusted metrics to scorecards



Inventory Planning

Prioritize high-turnover products; reduce slow-moving stock



Conclusion

15M+ Records Unified

Transactional data → vendor-centric analytical model

Scalable Dashboard

Power BI enables data-driven vendor & profitability decisions

Strategic Impact

Supports vendor selection, pricing, and inventory optimization